

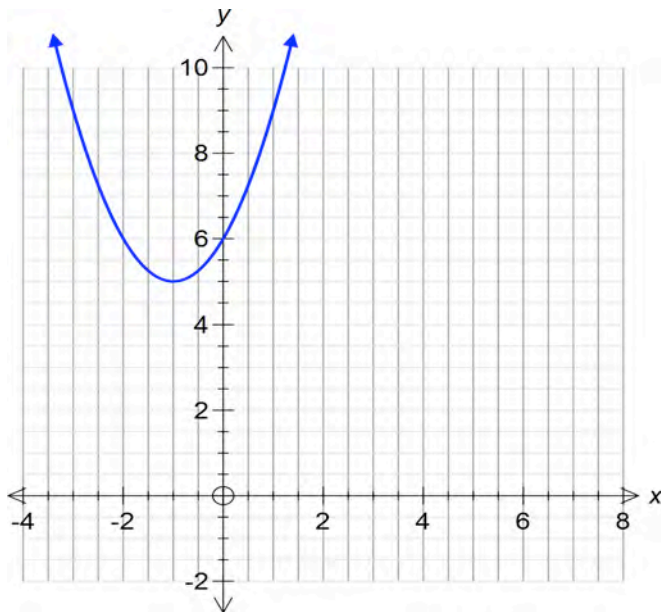
Question 1	(15 Marks)	<u>START QUESTION ON A NEW PAGE</u>	Marks
(a)	Write $4.0\dot{5}\dot{2}$ as a fraction in simplest terms		2
(b)	Simplify $\frac{3}{1+\sqrt{3}} - \frac{2}{1-\sqrt{3}}$		3
(c)	If the two shorter sides of a right angled triangle are $\sqrt{5}+1$ and $\sqrt{5}-1$, find the exact length of the hypotenuse		3
(d)	Factorise (i) $27x + x^4$ (ii) $9a^2 + 12ab + 4b^2$		4
(e)	Simplify $\frac{2a^2b+10ab}{(b+3)(b^2-3b+9)} \div \frac{a^2-25}{4(b+3)}$		3
Question 2	(12 Marks)	<u>START QUESTION ON A NEW PAGE</u>	
(a)	Solve (i) $\frac{2x}{3} - \frac{x-1}{2} = \frac{2}{9}$ (ii) $x = \frac{x+3}{x}$ (answer in surd form) (iii) $ 4r+1 = 1-r$		7
(b)	Solve $x^6 - 9x^3 - 8 = 0$		3
(c)	If $x = \frac{y+1}{y}$ express with y as the subject		2
Question 3	(12 Marks)	<u>START QUESTION ON A NEW PAGE</u>	
(a)	Solve the following Inequations: (i) $3-2x > 5$ (ii) $x^2 \leq 9$ (iii) $\frac{2x+1}{x} \geq 1$ (iv) $ x-3 < 1$		8

Question 3 Continued**Marks**

- (b) 4
- (i) Show that the graphs of $y = 2^x$ and $y = x + 2$ intersect at the points (0,2) and (2,4).
- (ii) Draw a neat sketch, indicating clearly the region represented by the intersection of the following inequalities: $y > 2^x$ and $y \leq x + 2$

Question 4 (20 Marks) START QUESTION ON A NEW PAGE

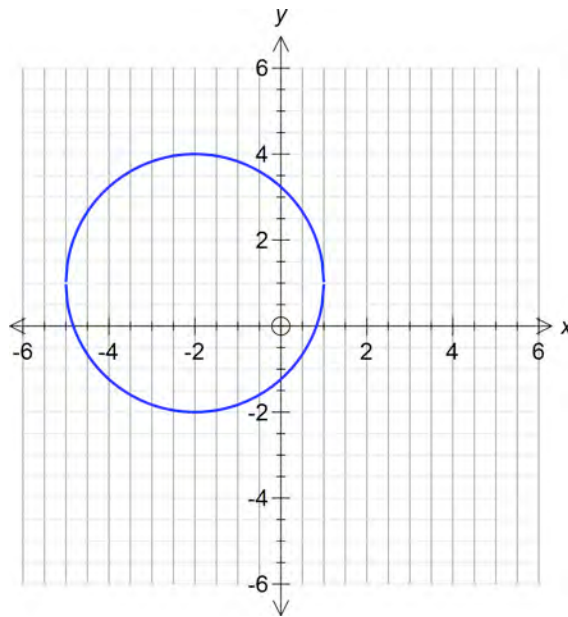
- (a) Sketch the graph of the function $f(x) = |2x + 1|$ clearly showing any intercepts with the axes. 2
- (b) For the graph below 5
- (i) State if it is a function
- (ii) Write down the domain and range
- (iii) On the copy of this graph attached to the end of this paper, sketch the inverse.



- (c) Find the inverse of the function $f(x) = 1 + \log_2 x$ 2
- (d) Sketch the graph of $y = 2 - \frac{1}{x+1}$ clearly showing clearly any asymptotes or intercepts with the axes. 3

Question 4 Continued**Marks**

- (e) Write down the equation of the circle below

2

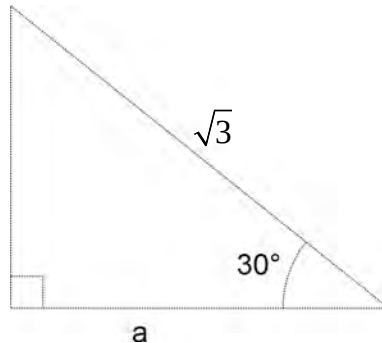
- (f) Consider the function $f(x) = \frac{-2x^2 + 1}{x^2 - 1}$ **6**
- (i) Show if the function is odd, even or neither.
 - (ii) Write down the equations of any vertical or horizontal asymptotes.
 - (iii) Find any intercepts
 - (iv) Draw a neat large sketch of this function showing all the above information clearly

Question 5 (12 Marks) START QUESTION ON A NEW PAGE

- (a) Solve the following trigonometric equations if $0^\circ \leq \theta \leq 360^\circ$: **9**
- (i) $\tan \theta = 1$
 - (ii) $\sin 2\theta = \frac{1}{2}$
 - (iii) $4\cos^2 \theta = \cos \theta$ (answer to the nearest minute where necessary)
- (b) If θ is obtuse and $\sin \theta = \frac{5}{7}$, without finding θ find the exact value of $\cos \theta$ **3**
and $\sec \theta$.

Question 6 (12 Marks) START QUESTION ON A NEW PAGE**Marks**

- (a) Find the exact value of a in the triangle below

2

- (b) Find the exact value of $\cot 330^\circ$

1

- (c) Prove that:

5

(i) $\cos^2(90^\circ - \theta) \cot \theta = \sin \theta \cos \theta$

(ii) $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$

- (d) Two ships A and B start from the same position P and sail in different directions. **4**

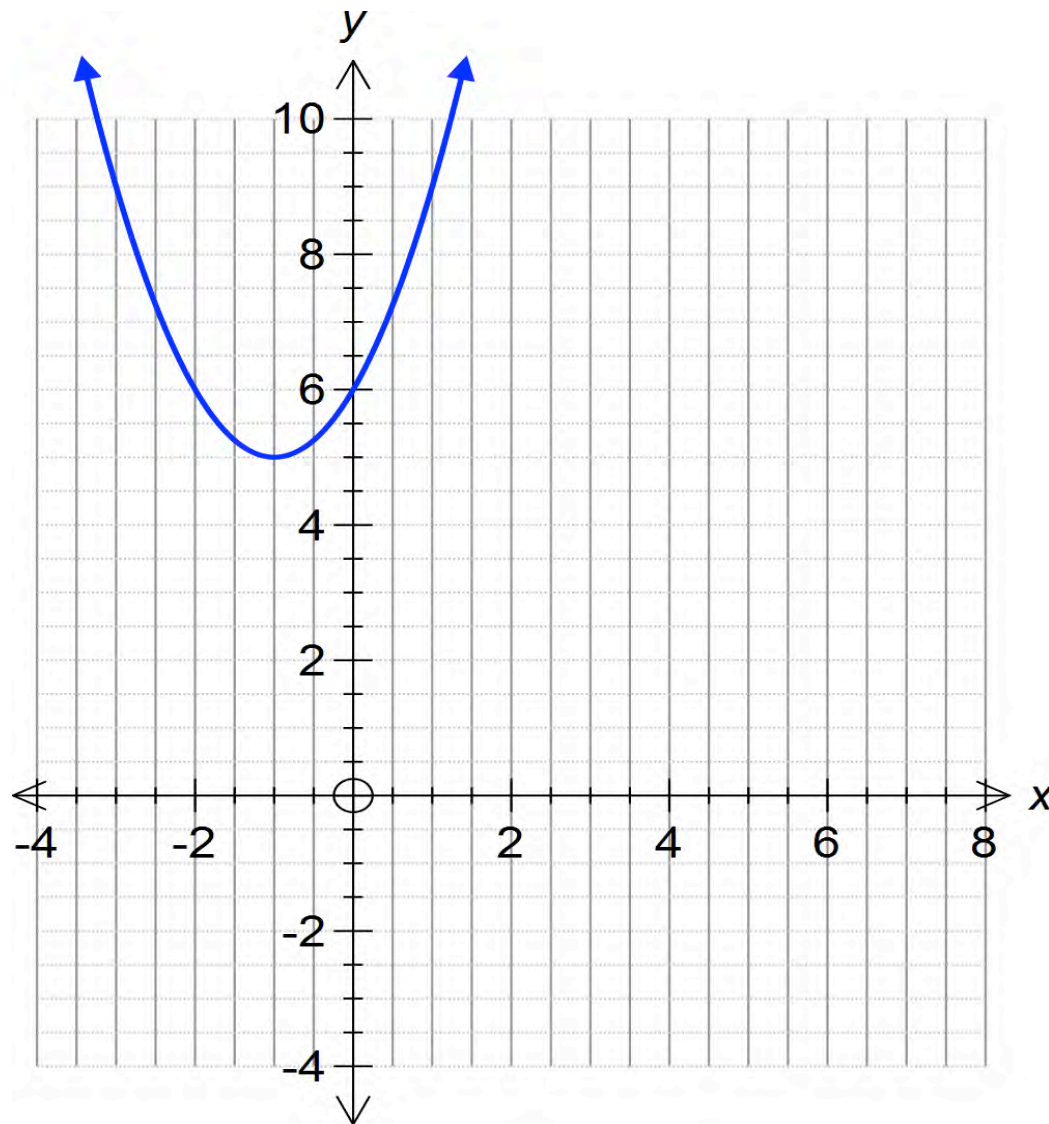
Ship A sails on a bearing of $350^\circ T$ at 23 knots. Ship B sails on a bearing of $260^\circ T$ at 18 knots. (note 1 knot is 1 nautical mile per hour)

- (i) Draw a neat diagram showing the path of both ships and their location after two hour's sailing.
- (ii) Show that $\angle APB = 90^\circ$
- (iii) If ship B remains stationary, find the bearing on which ship A will need to sail in order to rejoin ship B

END OF PAPER

Computer Number: _____

Use this sheet to answer Question 4 (b) (iii), hand in with the rest of Question 4.



Year 11 Ext 1

Question 1

(a) let $x = 4.05252...$

$10x = 40.5252...$ ✓ (1)

$1000x = 4052.5252...$ (2)

$990x = 4012$

$x = \frac{4012}{990}$ ✓

$x = 4 \frac{26}{495}$

$x = \frac{2006}{495}$

✗

1 mark either line 2 or 3

1 correct answer.

(in any way)

OR $x = 4.052525...$

$100x = 405.2525...$

$99x = 401.2$ ✓

(2) $x = \frac{401.2}{99}$

$x = \frac{4012}{990}$ ✓

(b) $\frac{3}{1+\sqrt{3}} - \frac{2}{1-\sqrt{3}}$

$= \frac{3(1-\sqrt{3})}{(1+\sqrt{3})(1-\sqrt{3})} - \frac{2(1+\sqrt{3})}{(1-\sqrt{3})(1+\sqrt{3})}$ ✓

$= \frac{3(1-\sqrt{3}) - 2(1+\sqrt{3})}{(1+\sqrt{3})(1-\sqrt{3})}$

$= \frac{3 - 3\sqrt{3} - 2 - 2\sqrt{3}}{1 - 3}$ ✓

both lines must be expanded correctly

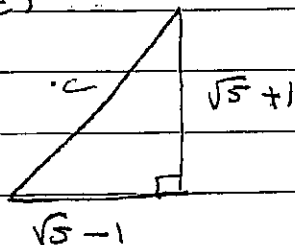
$= \frac{1 - 5\sqrt{3}}{-2}$

$= \frac{5\sqrt{3} - 1}{2}$ ✓

(3)

Question 1

(c)



$$c^2 = (\sqrt{5}+1)^2 + (\sqrt{5}-1)^2 \quad \checkmark$$

$$= 5 + 2\sqrt{5} + 1 + 5 - 2\sqrt{5} + 1 \quad \checkmark$$

$$c^2 = 12$$

$$c = \sqrt{12} \quad \checkmark$$

$$c = 2\sqrt{3} \quad \checkmark$$

(3)

(d)(i) $27x + x^4$

$$= x(27 + x^3) \quad \checkmark$$

$$= x(3+x)(9 - 3x + x^2) \quad \checkmark \quad (2)$$

(ii) $9a^2 + 12ab + 4b^2$

$$= (3a + 2b)(3a + 2b) \quad \checkmark \checkmark \quad (2)$$

$$= (3a + 2b)^2$$

(e) $\frac{2a^2b + 10ab}{(b+3)(b^2-3b+9)} \div \frac{a^2-25}{4(b+3)}$

$$= \frac{2ab(a+5)}{(b+3)(b^2-3b+9)} \times \frac{4(b+3)}{(a+5)(a-5)} \quad \checkmark$$

Fully factorised

$$= \frac{8ab}{(b^2-3b+9)(a-5)} \quad \checkmark \quad (3)$$

Frage 2

12

(a) (i) $\frac{2x}{3} - \frac{x-1}{2} = \frac{2}{9}$ (x18)

(2) $12x - 9(x-1) = 4$
 $12x - 9x + 9 = 4$

$3x = -5$
 $x = -\frac{5}{3}$

(ii) $x^2 = x + 3$ ($x \neq 0$)

(2) $x^2 - x - 3 = 0$
 $x = \frac{1 \pm \sqrt{1 - 4 \cdot 1 \cdot (-3)}}{2}$

$x = \frac{1 \pm \sqrt{13}}{2}$

(iii) $4r + 1 = 1 - r$

or $-(4r + 1) = 1 - r$

(3) $5r = 0$
 $r = 0$

$-4r - 1 = 1 - r$

$-3r = 2$
 $r = -\frac{2}{3}$

$-3r = 5$
 $r = -\frac{5}{3}$

Best solution: $r = 0$ is ok

$r = -\frac{5}{3}$ is invalid

b) $(x^3 - 8)(x^3 + 1) = 0$ | $(x^3 - 8)(x^3 - 1) = 0$

(3) $x^3 = 8$ $x^3 = -1$
 $x = 2$ $x = -1$

$x^3 = 8$ $x^3 = 1$
 $x = 2$ $x = 1$

$\Rightarrow xy = y + 1$

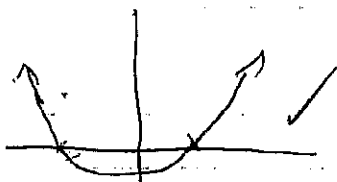
(2) $xy - y = 1$
 $y(x - 1) = 1$
 $y = \frac{1}{x - 1}$

Question 3

(a) (i) $3 - 2x > 5$
 $2x < -2$
 $x < -1$ ✓

①

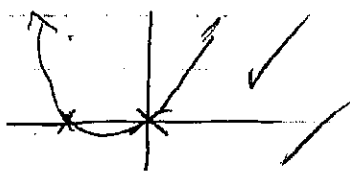
(ii) $x^2 \leq 9$
 $(x+3)(x-3) \leq 0$
 $-3 \leq x \leq 3$ ✓



②

(iii) $\frac{2x+1}{x} \geq 1$ ($x \neq 0$)

$x(2x+1) \geq x^2$
 $2x^2 + x - x^2 \geq 0$
 $x^2 + x \geq 0$
 $x(x+1) \geq 0$
 $x \leq -1$ or $x \geq 0$ ✓



③

(iv) $|x-3| < 1$

$x-3 < 1$ or $-x+3 < 1$
 $x < 4$ or $x > 2$
 $\therefore 2 < x < 4$ ✓

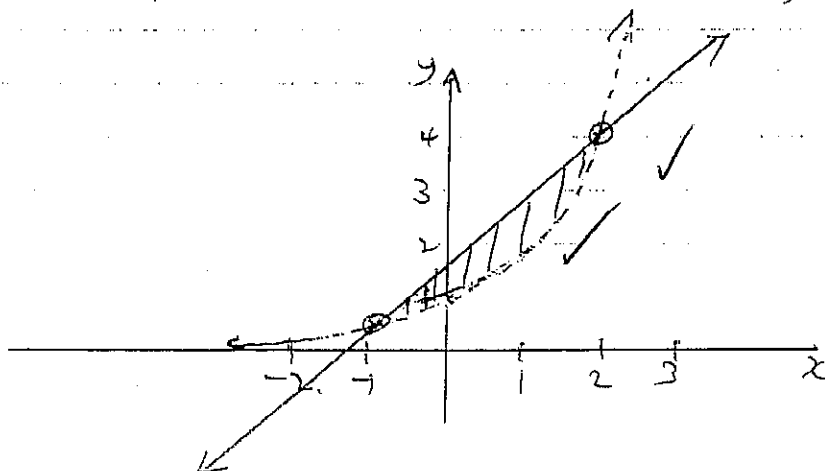
②

(b) $y = 2^x$ sub $x = -1$ $y = 2^{-1} = 1/2$
 $6y - 7x = 10$ $6y + 7 = 10, y = 1/2$ ✓

sub $x = 2$ $y = 2^2 = 4$
 $6y - 7 \times 2 = 10$
 $6y = 24$
 $y = 4$ ✓

②

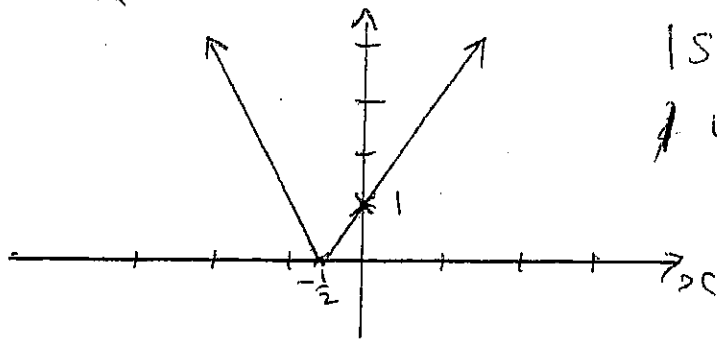
Since points lie on both lines they are points of intersection



1 for boundaries

1 for area. ②

question 4.



1 shape
1 intercept

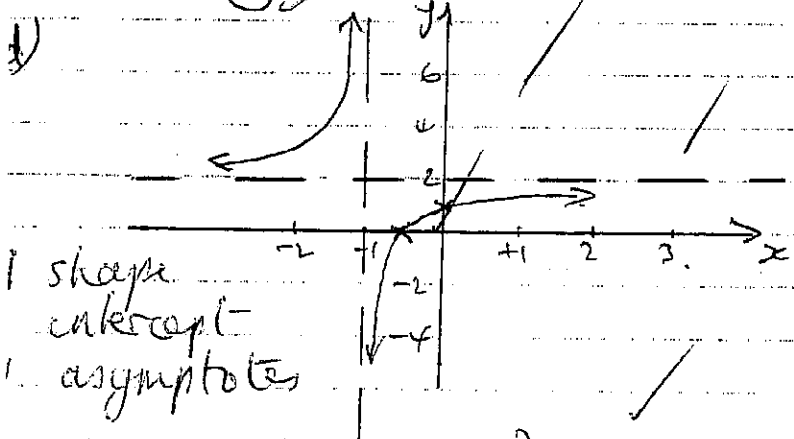
- (b) (i) yes ✓
(ii) domain $x \in \mathbb{R}$ ✓
range $y \geq 5$ ✓

(iii) see attached.

c) let $y = 1 + \log_2 x$
swap x, y for inverse
 $x = 1 + \log_2 y$
 $\log_2 y = x - 1$
 $y = 2^{x-1}$

let $y = \frac{x+1}{2x}$
 $x = \frac{y+1}{2y}$
 $2xy = y+1$

$2xy - y = 1$
 $y(2x-1) = 1$
 $y = \frac{1}{2x-1}$



$y - 2 = -\frac{1}{x+1}$

if $x = 0$ $y = 1$
if $y = 0$ $\frac{1}{x+1} = 2$
 $x = -\frac{1}{2}$

e) $(x+2)^2 + (y-1)^2 = 9$

1. $x^2 + y^2 = 9$

1. $(x+2)^2 + (y-1)^2 = \square$

$$f(-x) = \frac{-2(-x)^2 + 1}{(-x)^2 - 1}$$

$$= \frac{-2x^2 + 1}{x^2 - 1}$$

$$= f(x)$$

$\therefore$ even

$$(ii) f(x) = \frac{-2x^2 + 1}{(x+1)(x-1)}$$

$$\therefore x = \pm 1 \text{ are}$$

$$\text{also } f(x) = \frac{-\frac{2x^2}{x^2} + \frac{1}{x^2}}{\frac{x^2}{x^2} - \frac{1}{x^2}}$$

$$= \frac{-2 + \frac{1}{x^2}}{1 - \frac{1}{x^2}}$$

$$\text{as } x \rightarrow \infty \quad y \rightarrow -2$$

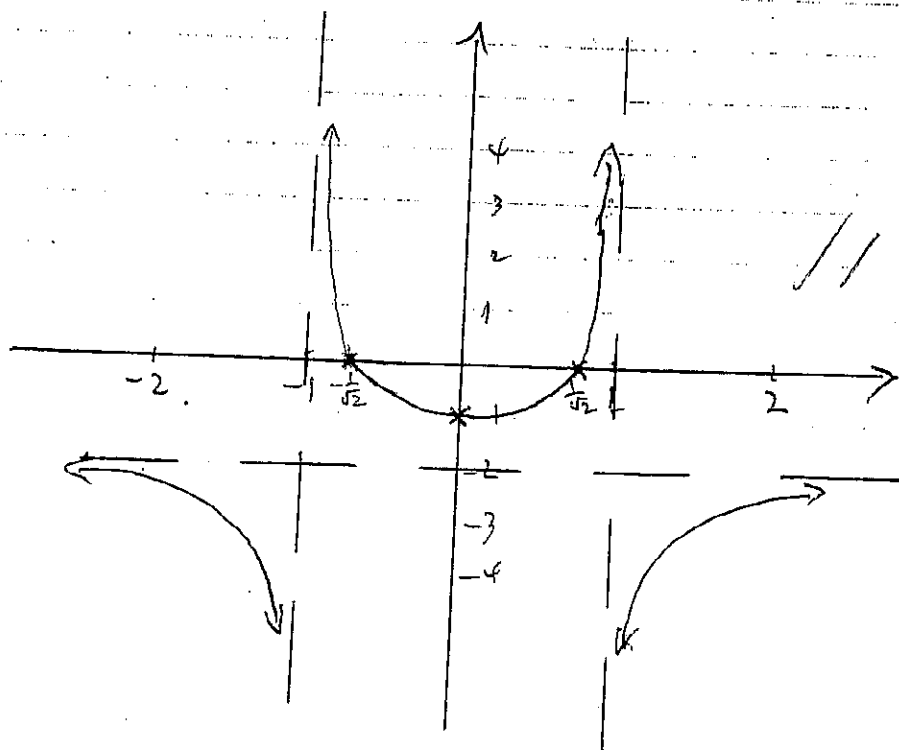
$\therefore y = -2$ is horizontal asymptote

$$i) \text{ when } x = 0, f(0) = -1$$

$$\text{when } f(x) = 0 \quad -2x^2 + 1 = 0 \quad \checkmark \quad (\text{both } x \text{ \& } y \text{ intercepts required})$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}} \quad (\pm 0.7)$$



// must make sense

Section 5

S/A
T/C

(i) $\tan \theta = 1$

$0 \leq \theta \leq 360$

related angle 45°

Quadrants 1, 4

$\therefore \theta = 45, 225$ ✓

or similar

(2)

(ii) $\sin 2\theta = -\frac{1}{2}$

$0 \leq 2\theta \leq 720$

related angle 30°

Quadrants 3, 4, 1, 2

~~$2\theta = 210, 330, 510, 690$~~ $2\theta = 30, 150, 390, 510$

~~$\theta = 105, 165, 255, 345$~~ $\theta = 15, 75, 195, 255$

3 marks - all solutions
2 marks - 2 solutions
1 mark - working
OK

(iii) $4 \cos^2 \theta - \cos \theta = 0$

$\cos \theta (4 \cos \theta - 1) = 0$ ✓

$\cos \theta = 0$ $\cos \theta = \frac{1}{4}$

$0 \leq \theta \leq 360$

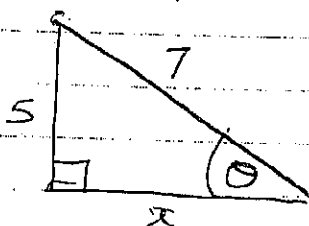
$\theta = 90, 270$ / related $< 75^\circ 31'$ ✓

Quadrants 1, 4

$\theta = 75^\circ 31', 284^\circ 29'$

$\therefore \theta = 90, 75^\circ 31', 270, 284^\circ 29'$

(6)



By Pythag $x = \sqrt{49 - 25}$
 $= \sqrt{24}$ ✓

$\cos \theta$ and $\sec \theta$ are negative in Q2 (obtuse angle)

$\therefore \cos \theta = -\frac{\sqrt{24}}{7}$ $\left(\frac{2\sqrt{6}}{7}\right)$ ✓

$\sec \theta = -\frac{7}{\sqrt{24}}$ $\left(\frac{7\sqrt{6}}{12}\right)$ ✓

[are mark if
 $\sec \theta = \frac{1}{\cos \theta}$ even if

$\cos \theta$ is incorrect)

Year 11 Ext 1 Q6

$$b(a) \cos 30^\circ = \frac{a}{\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\frac{1}{2} \cos 30^\circ$$

$$a = \frac{3}{2}$$

$\frac{2}{2}$ correct answer

$$(b) -\sqrt{3}$$

1

$$(c) (i) \text{ LHS} = \cos^2(90^\circ - \theta) \cot \theta$$

$$= \sin^2 \theta \cdot \frac{\cos \theta}{\sin \theta}$$

$$\frac{1}{2} \cos(90^\circ - \theta) = \sin \theta$$

or $\cot \theta = \frac{\cos \theta}{\sin \theta}$

$\frac{2}{2}$ correct solution

$$= \sin \theta \cos \theta$$

$$= \text{RHS}$$

$$(ii) \text{ LHS} = \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}$$

$\frac{1}{3}$ common denom:
correct

$$= \frac{\sin^2 \theta + 1 + 2 \cos \theta + \cos^2 \theta}{\sin^2 \theta (1 + \cos \theta)}$$

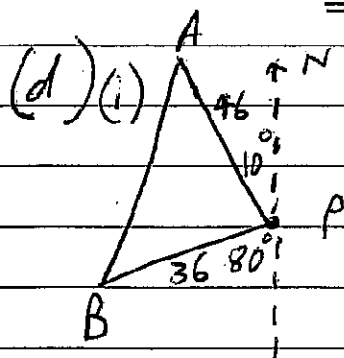
$\frac{2}{3}$ using $\sin^2 \theta + \cos^2 \theta = 1$

$$= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)}$$

$\frac{3}{3}$ correct solution

$$= 2 \operatorname{cosec} \theta$$

$$= \text{RHS}$$

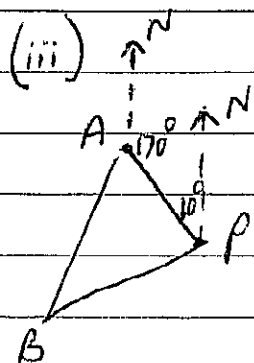


1 correct diagram

$$(ii) \angle APB = 180^\circ - (10^\circ + 80^\circ)$$

$$= 90^\circ$$

1 correct method



$$\tan \angle PAB = \frac{36}{46}$$

$$\frac{1}{2}$$

$$\angle PAB = 38^\circ \text{ (nearest degree)}$$

$$\text{Bearing} = 38^\circ + 170^\circ$$

$$\frac{2}{2}$$